# Soft Jellyfish Propulsion Simulation Proposal

Robert Marlow and Mark Lee

*Abstract*— **This Proposal explores the methods and existing literature for modeling soft robotic jellyfish propulsion. Through simulation of propulsion using a 2D, neutrally-buoyant, discrete elastic beam represented with a 5-node structure that replicates the contractile kinematics of a jellyfish bell, we aim to compare simulated performance metrics (forward speed, and thrust generation) to biological and robotic jellyfish studies.**

## I. Problem Statement and Motivation

Marine robots are becoming increasingly valuable to scientific study and commercial exploration, but remain limited in lifetime and range by propulsion sources that can be inefficient and hazardous to fragile sea-life [1]. As the field of soft robotics has matured, biomimetic designs including those inspired by jellyfish propulsion present a promising route to low-energy, highly maneuverable underwater platforms.

Jellyfish swim by producing coordinated bell contractions that generate vortex rings and jet thrust, achieving high efficiencies at low speeds [2]. This proposal seeks to evaluate the applicability of a discrete differential geometry model toward reproducing these fluid–structure interactions in simulation, which could decrease computational overhead for predicting the propulsive performance of soft robotic designs. Previous experimental and computational work on jellyfish-inspired robots and live-animal augmentation provides benchmarks and modeling motifs for our simulation-based exploration [4], [5].

## II. Initial Approach

We will start by modeling the jellyfish as a neutrally buoyant discrete elastic beam immersed in a viscous fluid. The simplest case to consider is a 2 dimensional system consisting of 5 nodes connected by stretching springs between adjacent nodes and bending springs between adjacent edges, which can then be scaled up by either increasing the node count in 2-dimensional space or by expanding to a 3-dimensional shell. By representing the natural curvature of the beam as a periodic function of time, we will emulate the bell contractions of a real jellyfish or jellyfish robot, then analyze its motion within the fluid.

Propulsive force will initially be abstracted as individual drag forces on the discretized nodes. This limited approach is expected to give an underestimate of actual propulsive force, as it considers neither the jetting effect that is a dominant factor in prolate species nor the vortex interactions that are dominant for oblate species [2]. If time permits, simulation accuracy will be improved by incorporating a more detailed model of the bell-fluid interaction.

## III. Background and Existing Literature

### A. Actuated Robotic Jellyfish: IPMC and SMA approaches

Najem et al. developed a biomimetic jellyfish-inspired underwater vehicle actuated by ionic polymer metal composites (IPMCs), demonstrating fabrication strategies and characterizing swimming performance for flexible-actuator-driven medusan robots. Their work shows that soft flexible actuators can reproduce bell-like contractions and serve as a practical benchmark for achievable kinematics in low-Reynolds number experimental platforms [2].

Earlier work at Virginia tech presented a jellyfish robot actuated by shape-memory-alloy (SMA) composite actuators. The robot, named "Robojelly" used SMA wires in parallel with spring steel spines to achieve large changes in natural curvature with magnitudes comparable to organic muscles. As shown in Figure 1, separate versions of the robot were built and characterized; one with a fully intact bell and one with "segmented" flaps around the bell margin. These impact of this change in bell shape could also be a target for modeling by our simulation. The study further emphasizes fabrication, kinematics replication, and experimental measurements of swimming performance, providing additional validation points for actuator timing and amplitude in biomimetic designs [1].

Fig. 1. Images of Robojelly with unsegmented (top) and segmented (bottom) bell margin [1].

### B. Mechanical Design and Motion Control

Although it revolves around a mostly rigid robot, the work presented at the Chinese Control Conference (Li & Yu) discusses mechanical structure-driven robotic jellyfish designs, motion analysis, and simulation of control schemes for steering and thrust modulation. An actuated ballast was incorporated to vary the center of the bell for bioinspired orientation control [3]. This study indicates the value of coupling mechanical design choices to control strategies to achieve desired swimming behaviors and maneuverability in robotic jellyfish.

### C. Computational FSI and Morphological Effects

Miles & Battista used IB2d, an open-source implementation of the immersed boundary mathematical framework, to study how tentacles and oral arms affect locomotion and found that appendages can strongly influence vortex formation and forward speed. In many parameter regimes, increased density or length of tentacles or oral arms were found to significantly decrease forward speed and change energy dissipation/mixing behavior [4].

### D. Biological Augmentation and Application Context

Recent work by Anuszczyk & Dabiri demonstrates electromechanical stimulation of live jellyfish muscle tissue for ocean exploration applications, showing the potential for hybrid bio-robotic systems and reinforcing the broader interest in jellyfish-inspired platforms for low-power ocean monitoring. Their discussion provides context for realistic performance goals and operational use-cases for jellyfish-inspired vehicles [5].

## IV. Planned Experiments and Metrics

- **2D simulation:** Run cycles of prescribed bell contraction/relaxation and measure mean forward velocity and impulse per stroke.
- **Parameter experimentation:** Vary stroke amplitude, contraction duration, fineness ratio (undeformed geometry), bell stiffness, and Reynolds number to map performance.

## V. Conclusion

The goal of our simulation is to investigate how oscillatory bell movement propels a soft jellyfish in a viscous fluid. To achieve this, we propose simulation approach based on discrete differential geometry, beginning with a 5-node elastic beam that can then be expanded to add more nodes in 2 dimensions, and potentially a 3-dimensional shell. A simplified approach to evaluating fluid forces will be implemented, with room for expansion if necessary. Model validation will be informed by approaches and findings from existing literature, including testing of soft jellyfish robot prototypes, testing of biological jellyfish, and mathematical models of fluid-structure interaction. By varying material properties, geometric configurations, and control signal profiles, we aim to examine how these parameters influence propulsion and assess the model's suitability for future robotic design applications.

## References


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